1 **(a)** $\begin{pmatrix}12\\-9\end{pmatrix}$ or $\begin{pmatrix}-4\\10\end{pmatrix}$ seen **[M1]**

$\begin{pmatrix}16\\-19\end{pmatrix}$ **[A1]**

(b) $\vec{AB} = \begin{pmatrix}2\\2\end{pmatrix}$ **[M1]**

$(5, 1)$ **[A1]**

[Total Marks: 4]

2 **(a)** $\vec{AB} = \mathbf{b} - \mathbf{a}$ or $\vec{BA} = \mathbf{a} - \mathbf{b}$ **[B1]**

$\vec{OP} = \mathbf{a} + \frac{3}{4}(\mathbf{b} - \mathbf{a})$ or $\vec{OP} = \mathbf{b} + \frac{1}{4}(\mathbf{a} - \mathbf{b})$ **[M1]**

$\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

(b) $\vec{MP} = -\frac{1}{2}\mathbf{a} + \vec{OP}$ (oe) **[M1]**

$-\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

[Total Marks: 5]

3 **(a)** $\vec{AX} = \frac{1}{3}\mathbf{c}$ **[M1]**

$\mathbf{a} + \frac{1}{3}\mathbf{c}$ **[A1]**

(b) $\vec{BY} = 2\mathbf{a}$ or $\vec{CY} = 3\mathbf{a}$ **[M1]**

$3\mathbf{a} + \mathbf{c}$ **[A1]**

(c) $\vec{OY} = 3\mathbf{a} + \mathbf{c} = 3(\mathbf{a} + \frac{1}{3}\mathbf{c}) = 3\vec{OX}$ **[M1]**

Since $\vec{OY}$ is a scalar multiple of $\vec{OX}$ and they share the common point O, they are collinear. **[A1]**

(d) $1 : 2$ **[B1]**

[Total Marks: 7]

4 **(a)** $3 \times \begin{pmatrix}4\\-3\end{pmatrix}$ **[M1]**

$\begin{pmatrix}12\\-9\end{pmatrix}$ **[A1]**

(b) $\begin{pmatrix}4\\-3\end{pmatrix} - \begin{pmatrix}-4\\10\end{pmatrix}$ or $\begin{pmatrix}4-2(-2)\\-3-2(5)\end{pmatrix}$ **[M1]**

$\begin{pmatrix}8\\-13\end{pmatrix}$ **[A1]**

(c) $\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$ or $(6-(-2), -1-5)$ **[M1]**

$(8, -6)$ **[A1]**

[Total Marks: 6]

5 (a) $\begin{pmatrix} 7 \\ 10 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ 12 \end{pmatrix}$ **[A1]**

(b) $\vec{AC} = \begin{pmatrix} 4-1 \\ 4-(-2) \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ **[M1]**

$\frac{1}{2}\vec{AB} = \frac{1}{2}\begin{pmatrix} 6 \\ 12 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, hence $\vec{AC} = \frac{1}{2}\vec{AB}$ **[A1]**

(c) $(4-5, 4+2)$ **[M1]**

$(-1, 6)$ **[A1]**

(d) $\begin{pmatrix} -1 \\ 6 \end{pmatrix} - \begin{pmatrix} 7 \\ 10 \end{pmatrix}$ or equivalent path method $-\vec{AC} + \vec{CD}$ **[M1]**

$\begin{pmatrix} -8 \\ -4 \end{pmatrix}$ **[A1]**

[Total Marks: 8]

6 (a) $\vec{PQ} = 2\begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ -8 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ -8 \end{pmatrix}$ or $(5+7, -1-8)$ **[M1]**

$(12, -9)$ **[A1]**

(b) $\vec{PR} = \begin{pmatrix} 14-5 \\ 3-(-1) \end{pmatrix} = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$ **[B1]**

$3h - k = 9$ and $-2h + 4k = 4$ **[M1]**

Rearranging to get $k = 3h - 9$ and substituting: $-2h + 4(3h - 9) = 4$ or equivalent valid method to solve simultaneous equations **[M1]**

$10h = 40 \implies h = 4$ **[A1]**

$k = 3(4) - 9 = 3$ **[A1]**

[Total Marks: 8]

7 (a) $\begin{pmatrix} 8 \\ -6 \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ -1 \end{pmatrix}$ **[A1]**

(b) $(5, 1)$ **[B1]**

[Total Marks: 3]

8 (a) Forms a correct equation for the horizontal components: $3(4) - k = 10$ **[M1]**

$k = 2$ **[A1]**

Forms a correct equation for the vertical components: $3(-3) + 2y = 6$

-9 + 2y = 6$[M1]$

$y = 7.5$ **[A1]**

(b) (i) $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$ **[B1]**

(ii) Finds vector $\vec{QR} = \begin{pmatrix} 6 \\ 12 \end{pmatrix}$ OR $\vec{PR} = \begin{pmatrix} 9 \\ 18 \end{pmatrix}$ **[M1]**

States that $\vec{QR} = 2\vec{PQ}$ (or equivalent scalar multiplier), showing the vectors are parallel, AND mentions that they share a common point Q (or P), concluding they are collinear **[A1]**

[Total Marks: 6]

9 (a) $\begin{pmatrix} 8 \\ -1 \end{pmatrix}$ **[B1]**

(b) Deduces correct column vectors for **a** and **b**: $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ **[B1]**

Sets up correct simultaneous equations from components: $2p - q = 8$ and $p + 2q = -1$ **[M1]**

Valid method to solve the simultaneous equations (e.g. multiplying first by 2 and adding: $4p - 2q = 16$

5p = 15)$[M1]$

$p = 3$ **[A1]**

$q = -2$ **[A1]**

[Total Marks: 5]